

```
#count duplicates:  
print(df.duplicated().sum())
```

0

#5. Dropping duplicate records

#Before

```
print(df.shape)
```

#Remove duplicates

```
df.drop_duplicates(inplace=True)
```

#After

```
print(df.shape)
```

(52, 6)

(52, 6)

#6. String operations

#string methods work only on text col

#Remove extra spaces

```
df['Student_Name'] = df['Student_Name'].str.strip() #using strip to remove spaces
```

#convert to uppercase

```
df['Student_Name'] = df['Student_Name'].str.upper()
```

```
print(df['Student_Name'].head())
```

#Convert to title case

```
df['Student_Name'] = df['Student_Name'].str.title() #1st letter only caps
```

```
print(df['Student_Name'].head())
```

0 STUDENT1

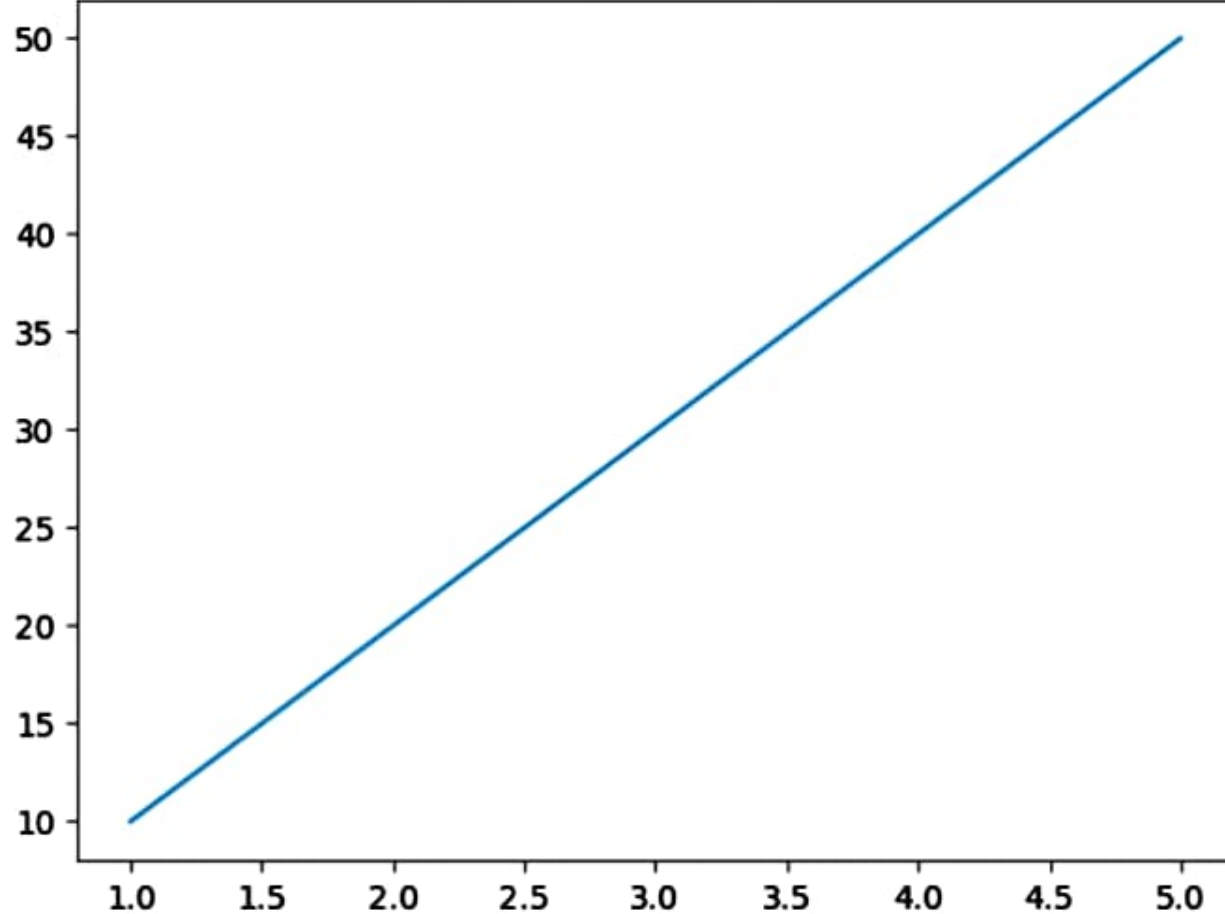
```
2      Electrical and Electronic  
3      Mechanical  
4      Computer Science  
Name: Department, dtype: object
```

```
: #String Length  
df['Name_Length']=df['Student_Name'].str.len() #add new col and find len  
print(df[['Student_Name','Name_Length']].head())
```

	Student_Name	Name_Length
0	Student1	8
1	Student2	8
2	Student3	8
3	Student4	8
4	Student5	8

```
: #Extract first character  
df['First_Letter']=df['Student_Name'].str[0] #access val  
  
print(df[['Student_Name','First_Letter']].head())
```

	Student_Name	First_Letter
0	Student1	S
1	Student2	S
2	Student3	S
3	Student4	S
4	Student5	S



```
#Figure and Axes
#These are the most imp concepts
#Figure-The entire chart window
#Axes-The area where the graph is drawn
import matplotlib.pyplot as plt
fig,ax=plt.subplots() #empty subplot means 1 chart,if mention 4 ,4 charts create
ax.plot([1,2,3],[10,20,30])
#Adding a title
plt.title("My Plot")
```

```
!pip install matplotlib
```

```
Requirement already satisfied: matplotlib in c:\users\dell-anaconda3\lib\site-packages (3.9.2)  
Requirement already satisfied: contourpy>=1.0.1 in c:\users\dell-anaconda3\lib\site-packages (from matplotlib) (1.2.0)  
Requirement already satisfied: cyclor>=0.10 in c:\users\dell-anaconda3\lib\site-packages (from matplotlib) (0.11.0)  
Requirement already satisfied: fonttools>=4.22.0 in c:\users\dell-anaconda3\lib\site-packages (from matplotlib) (4.51.0)  
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\dell-anaconda3\lib\site-packages (from matplotlib) (1.4.4)  
Requirement already satisfied: numpy>=1.23 in c:\users\dell-anaconda3\lib\site-packages (from matplotlib) (1.26.4)  
Requirement already satisfied: packaging>=20.0 in c:\users\dell-anaconda3\lib\site-packages (from matplotlib) (24.1)  
Requirement already satisfied: pillow>=8 in c:\users\dell-anaconda3\lib\site-packages (from matplotlib) (10.4.0)  
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\dell-anaconda3\lib\site-packages (from matplotlib) (3.1.2)  
Requirement already satisfied: python-dateutil>=2.7 in c:\users\dell-anaconda3\lib\site-packages (from matplotlib) (2.9.0.post0)  
Requirement already satisfied: six>=1.5 in c:\users\dell-anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib) (1.16.0)
```

```
import matplotlib.pyplot as plt #only basic use
```

```
x=[1,2,3,4,5]
```

```
y=[10,20,30,40,50]
```

```
plt.plot(x,y) #it returns line
```

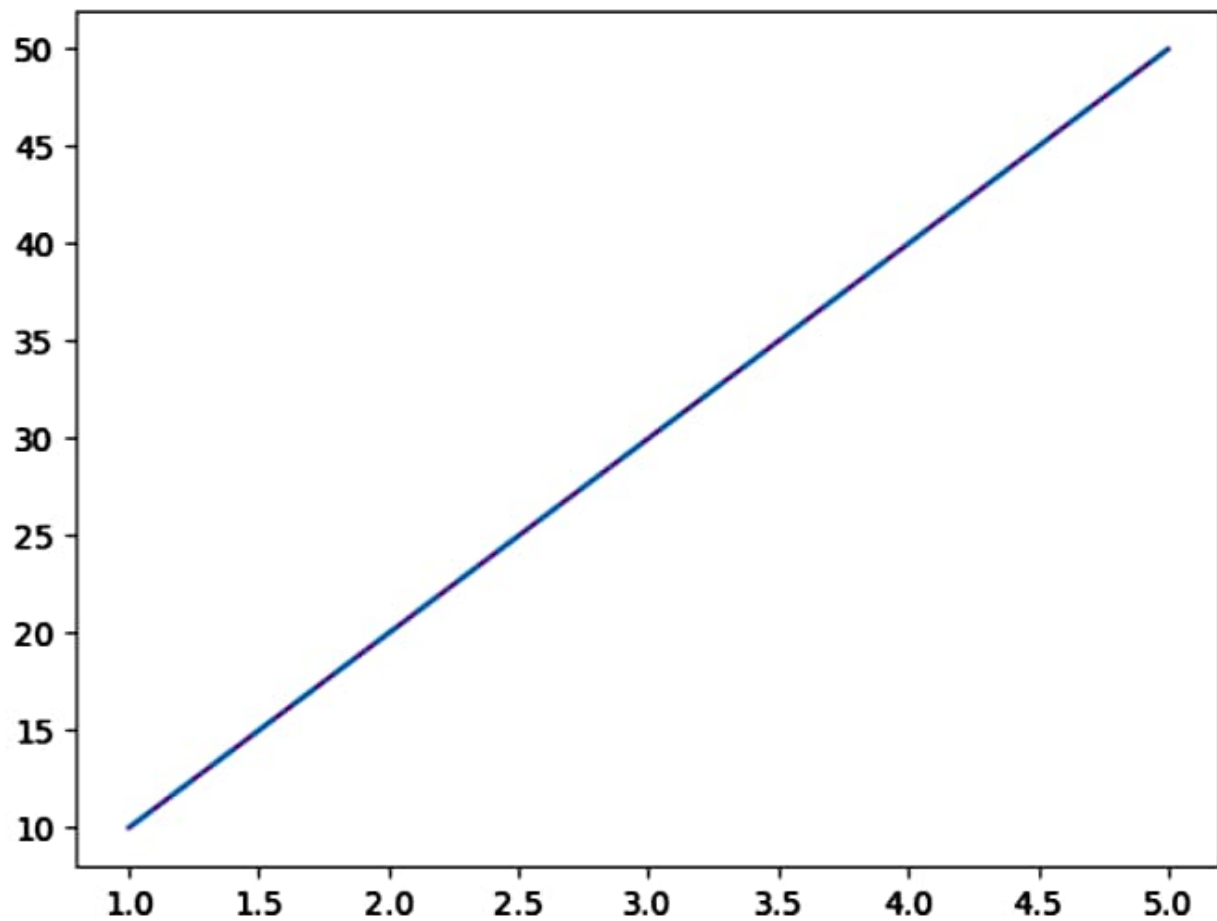
```
plt.show() #instead of print here we are using show
```

```
fig,ax=plt.subplots(1,1) #empty subplot means 1 chart, 1 dimension + , 1 chart is created
ax.plot([1,2,3],[10,20,30])
#Adding a title
plt.title("Monthly Sale")
#adding x and y labels
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()

#changing line color
plt.plot(x,y,color='purple')
#Changing Line Style
plt.plot(x,y,linestyle='-.')
plt.show()
```



Month



```
!pip install seaborn
```

```
Requirement already satisfied: seaborn in c:\users\dell-anaconda3\lib\site-packages (0.13.2)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in c:\users\dell-anaconda3\lib\site-packages (from seaborn) (1.26.4)
Requirement already satisfied: pandas>=1.2 in c:\users\dell-anaconda3\lib\site-packages (from seaborn) (2.2.2)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in c:\users\dell-anaconda3\lib\site-packages (from seaborn) (3.9.2)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\dell-anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.2.0)
Requirement already satisfied: cycler>=0.10 in c:\users\dell-anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\dell-anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (4.51.0)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\dell-anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.4.4)
Requirement already satisfied: packaging>=20.0 in c:\users\dell-anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (24.1)
Requirement already satisfied: pillow>=8 in c:\users\dell-anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (10.4.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\dell-anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (3.1.2)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\dell-anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\dell-anaconda3\lib\site-packages (from pandas>=1.2->seaborn) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\dell-anaconda3\lib\site-packages (from pandas>=1.2->seaborn) (2023.3)
Requirement already satisfied: six>=1.5 in c:\users\dell-anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->seaborn) (1.16.0)
```

```
import seaborn as sns #advance level ,attractive
import matplotlib.pyplot as plt

x=[1,2,3,4,5]
y=[10,20,30,40,50]

#Themes in seaborn
#Themes control the appearance of charts.
#default theme
sns.set_theme()
#dark grid theme
sns.set_style("dark")
sns.lineplot(x=x,y=y)
plt.show()
```



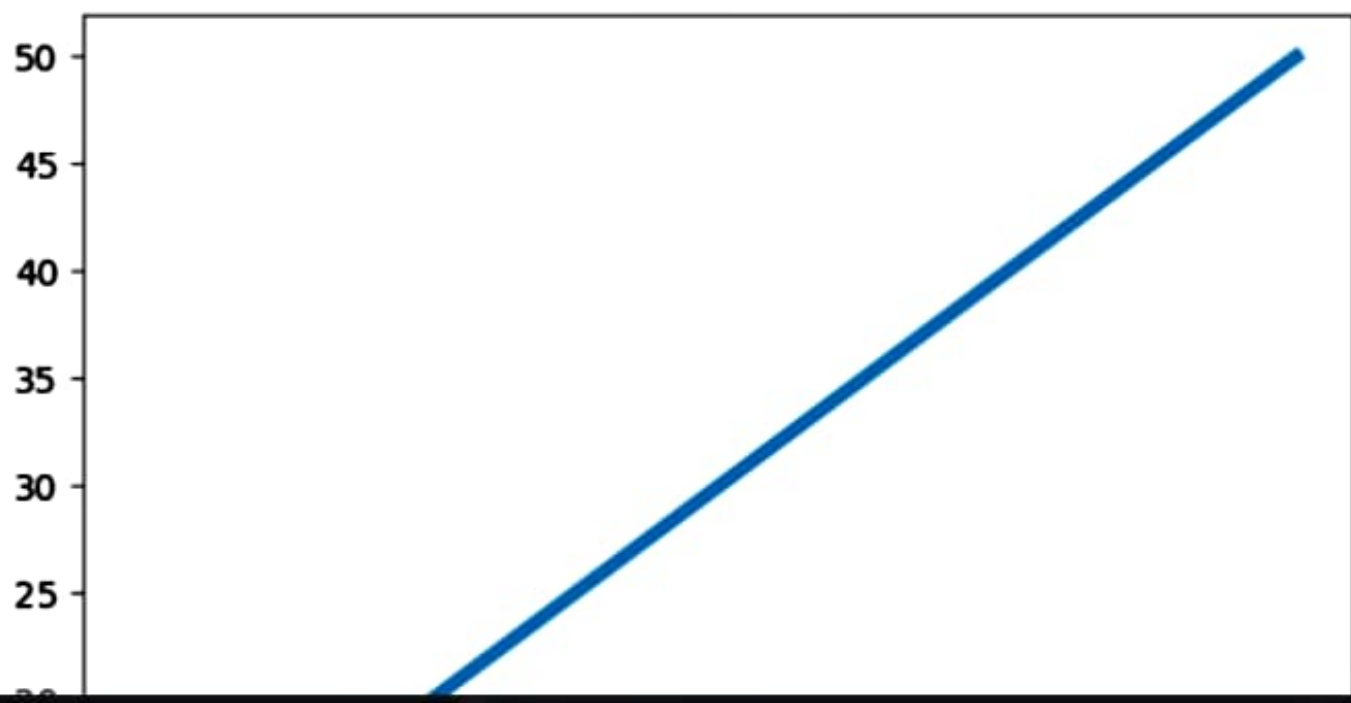
```
#Changing Line Width
plt.plot(x,y,linewidth=4) #line broad
plt.show()
#Thicker line.

#Adding Markers
plt.plot(x,y,marker='s')

plt.show()

#Output:
#.-.-.-.-.-.
#Marker types:

#'o' Circle
#'s' Square
#'^' Triangle
#'*' Star
```




```
sns.set_style("white")
sns.lineplot(x=x,y=y)
plt.show()
```

```
sns.set_style("ticks")
sns.lineplot(x=x,y=y)
plt.show()
```

```
sns.set_style("darkgrid")
sns.lineplot(x=x,y=y)
plt.show()
```

#Available styles

#darkgrid

#whitegrid

#dark

#white

#ticks #white and tick is same white

```
: #Color palettes
#A palette is a collection of colors

#View current palette
sns.set_palette("bright") #bright the blue color, if we put deep it dim the line .box is there bcz previous i mentioned darkgrid

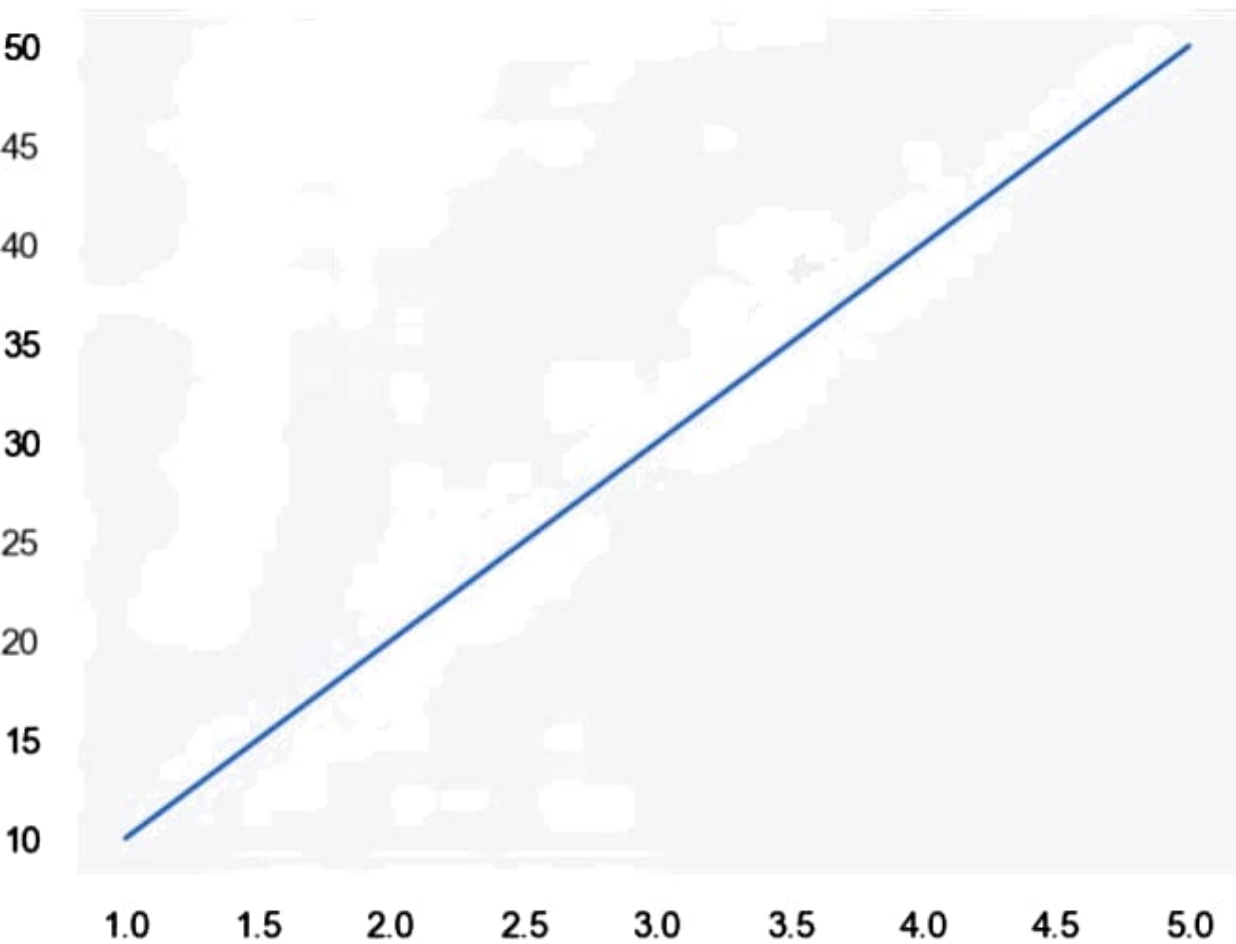
sns.lineplot(
    x=[1,2,3,4],
    y=[10,20,30,40]
)
plt.show()

sns.set_palette("pastel")
sns.lineplot(
    x=[1,2,3,4],
    y=[10,20,30,40]
)
plt.show()

sns.set_palette("colorblind")

sns.lineplot(
    x=[1,2,3,4],
    y=[10,20,30,40]
)
plt.show()

#available style
#deep
#bright
#muted
```



#available style

#deep

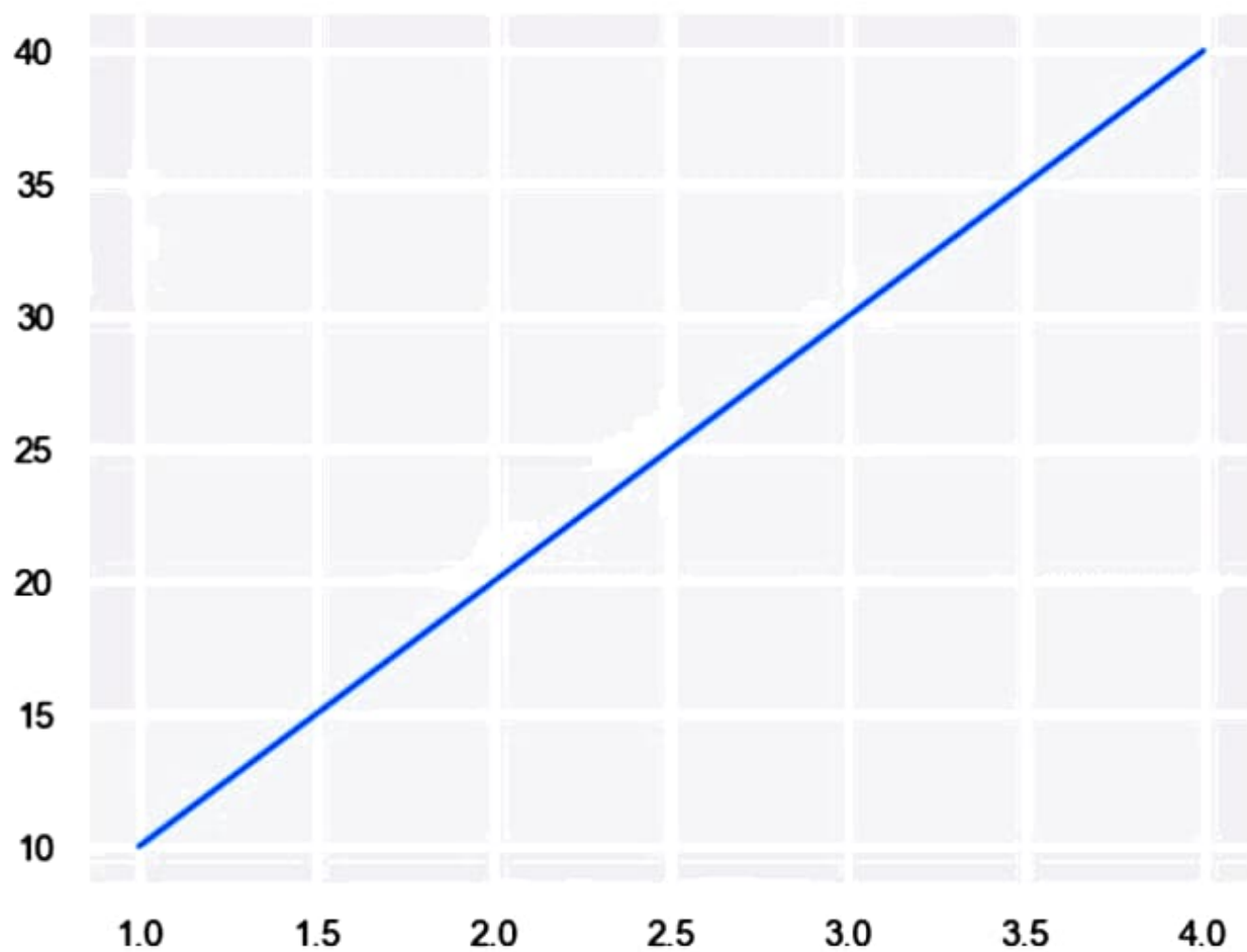
#bright

#muted

#pastel

#dark

#colorblind



```
#Built-in dataset support
#One of seaborn's biggest advantages is built-in datasets
#View available datasets
import seaborn as sns
print(sns.get_dataset_names())
```

```
['anagrams', 'anscombe', 'attention', 'brain_networks', 'car_crashes', 'diamonds', 'dots', 'dowjones', 'exercise', 'flights', 'fmri', 'geyser', 'glue', 'healthexp', 'iris', 'mpg', 'penguins', 'planets', 'seaice', 'taxis', 'tips', 'titanic']
```

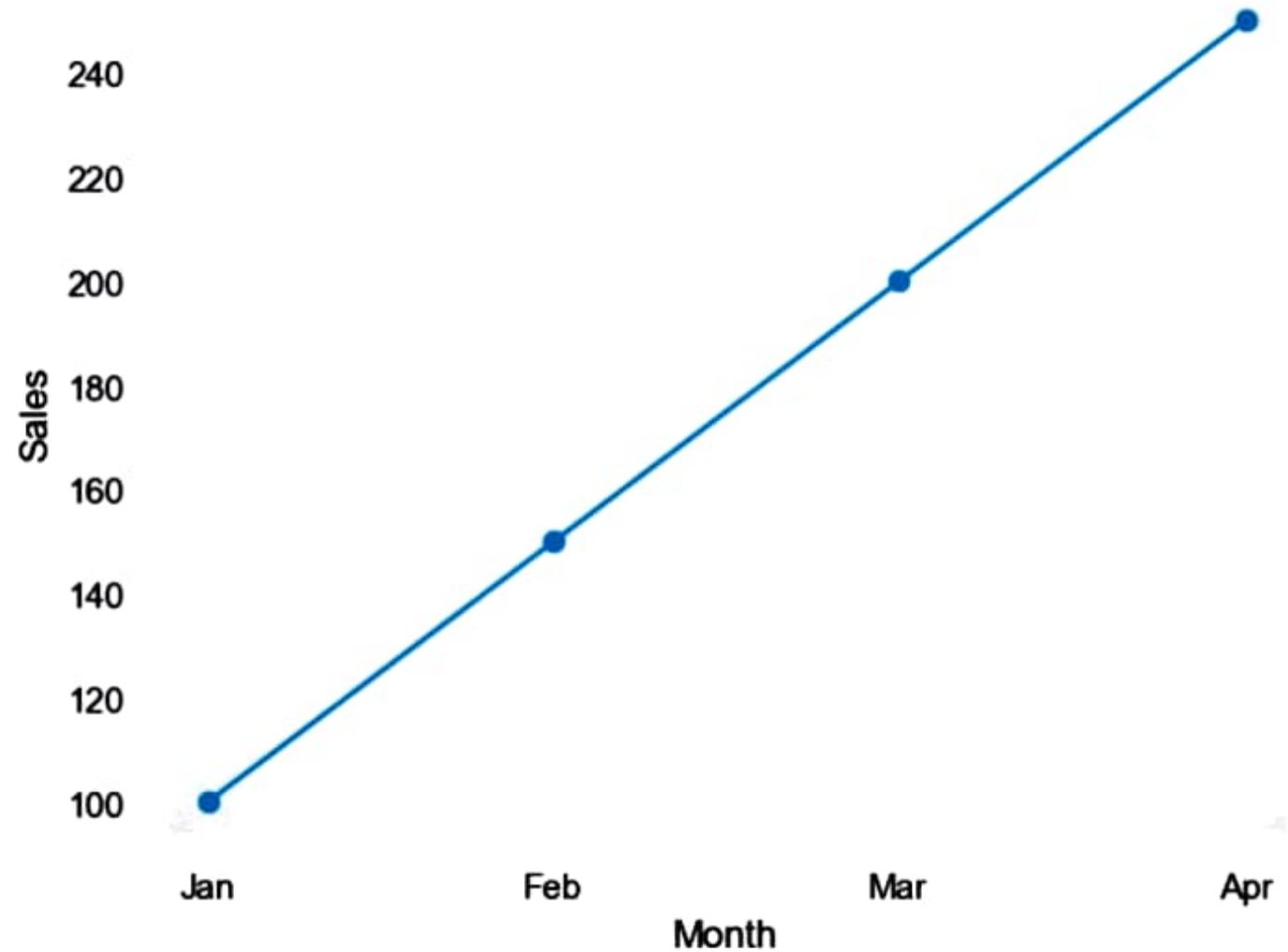
```
#Load titanic dataset
import seaborn as sns
df=sns.load_dataset("titanic")
print(df.head())
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class \
0	0	3	male	22.0	1	0	7.2500	S	Third
1	1	1	female	38.0	1	0	71.2833	C	First
2	1	3	female	26.0	0	0	7.9250	S	Third
3	1	1	female	35.0	1	0	53.1000	S	First
4	0	3	male	35.0	0	0	8.0500	S	Third

	who	adult_male	deck	embark_town	alive	alone
0	man	True	NaN	Southampton	no	False
1	woman	False	C	Cherbourg	yes	False
2	woman	False	NaN	Southampton	yes	True
3	woman	False	C	Southampton	yes	False
4	man	True	NaN	Southampton	no	True

```
#Line chart
#what is a line chart?
```

Monthly Sales



#when to use?
#Showing trends ove time
#Examples:
#-Monthly sales
#-Temperature changes
#-Stock price
#-Student performance over semesters

```
import matplotlib.pyplot as plt

months=['Jan','Feb','Mar','Apr']
sales=[100,150,200,250]

plt.plot(months,sales,marker='o')

plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()
```

```
#Bar chart
#what is a bar chart?
#A bar chart uses rectangular bars to compare values.
```

```
#When to use
```

```
#Comparing categories
```

```
#Examples:
```

```
#-Dept wise employees
```

```
#-Product sales
```

```
#-Student marks
```

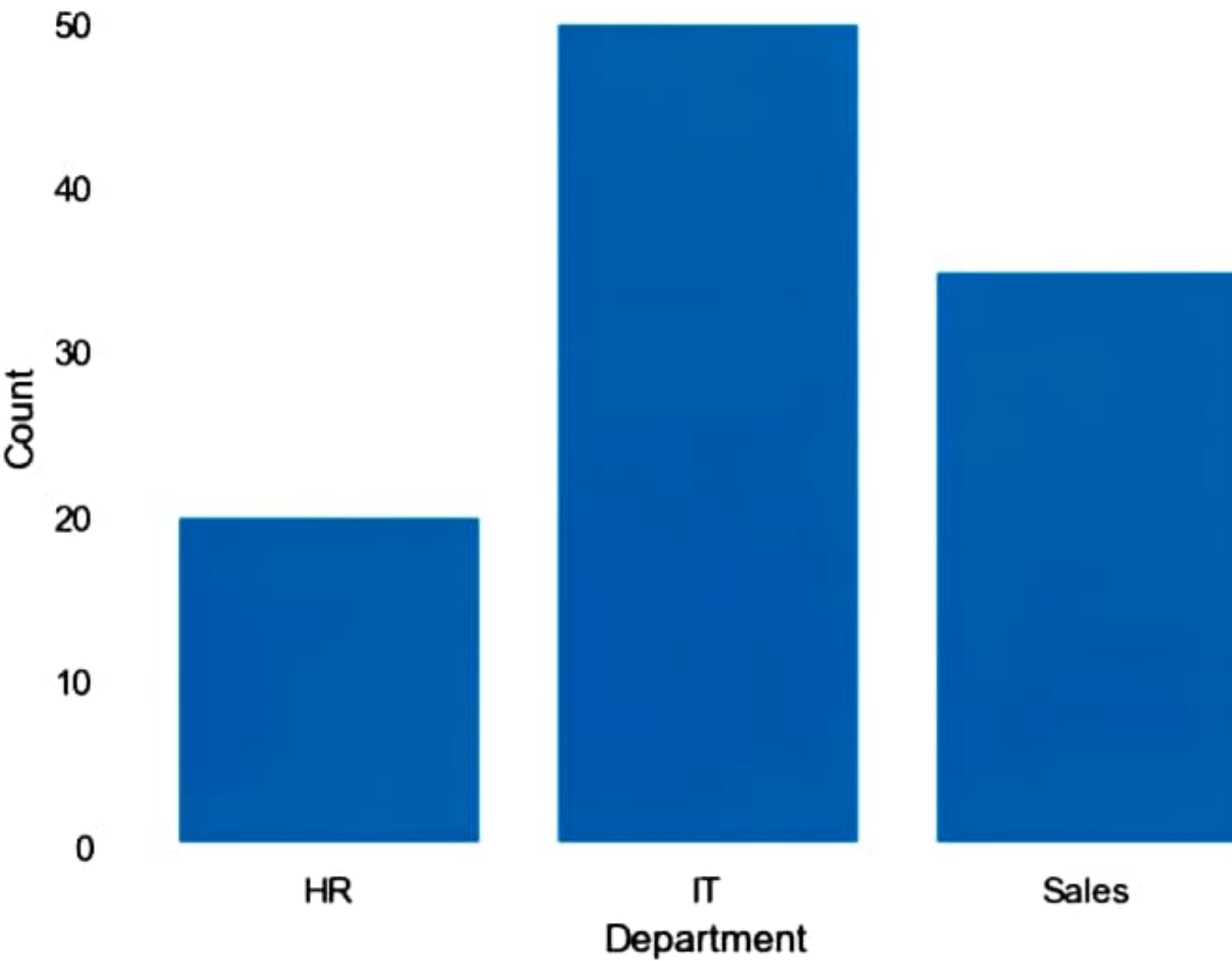
```
#-Population by city
```

```
import matplotlib.pyplot as plt
dept=['HR','IT','Sales']
employees=[20,50,35]
```

```
plt.bar(dept,employees)
```

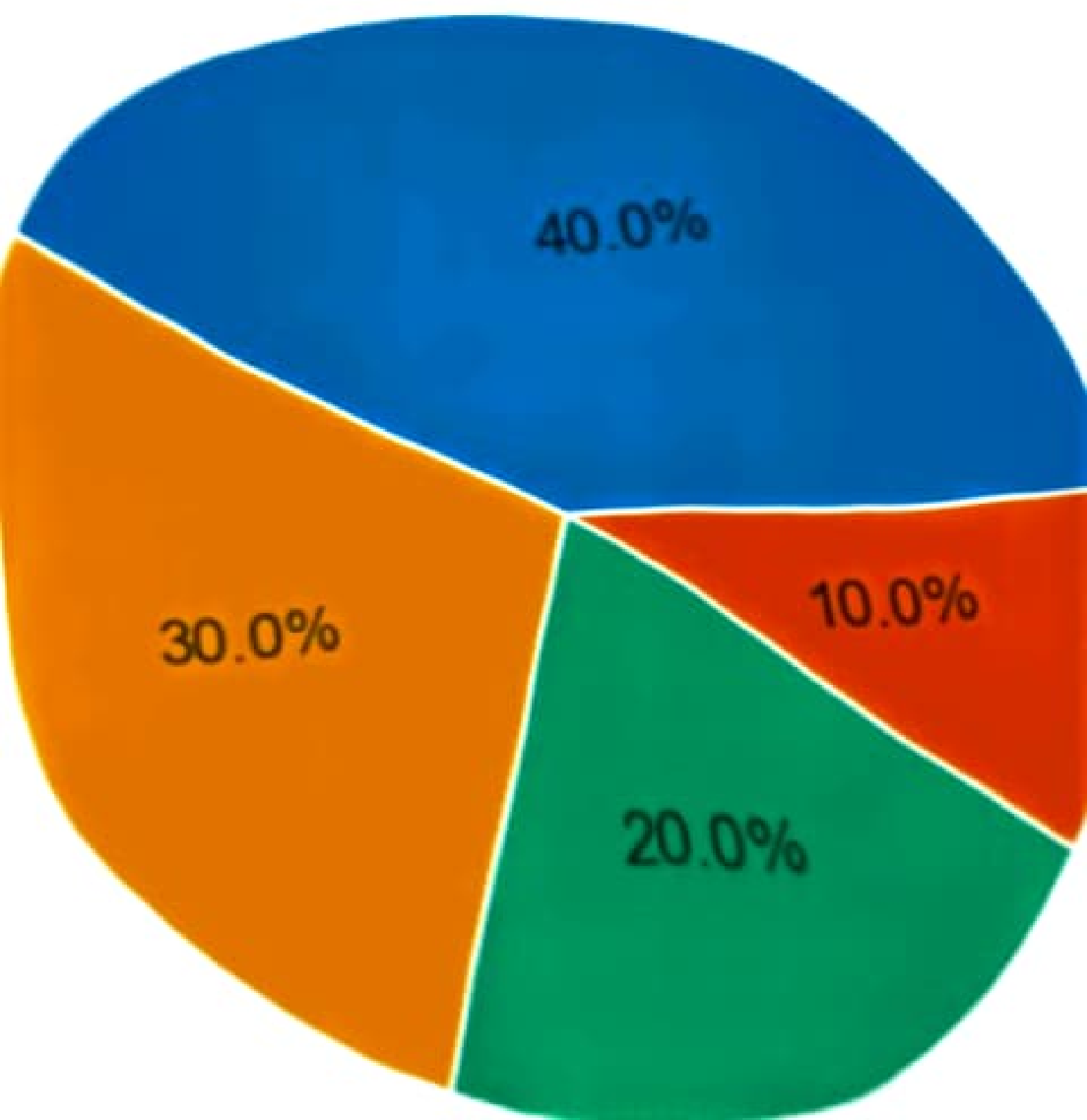
```
plt.title("Employees by Department")
plt.xlabel("Department")
plt.ylabel("Count")
plt.show()
```


Employees by Department



Monthly expenses

Food



#Pie chart

#What is a pie chart?

#A piechart shows parts of a whole

#When to use

#Showing percentages

#Examples:

#-Budget allocation

#-Market share

#-Survey results

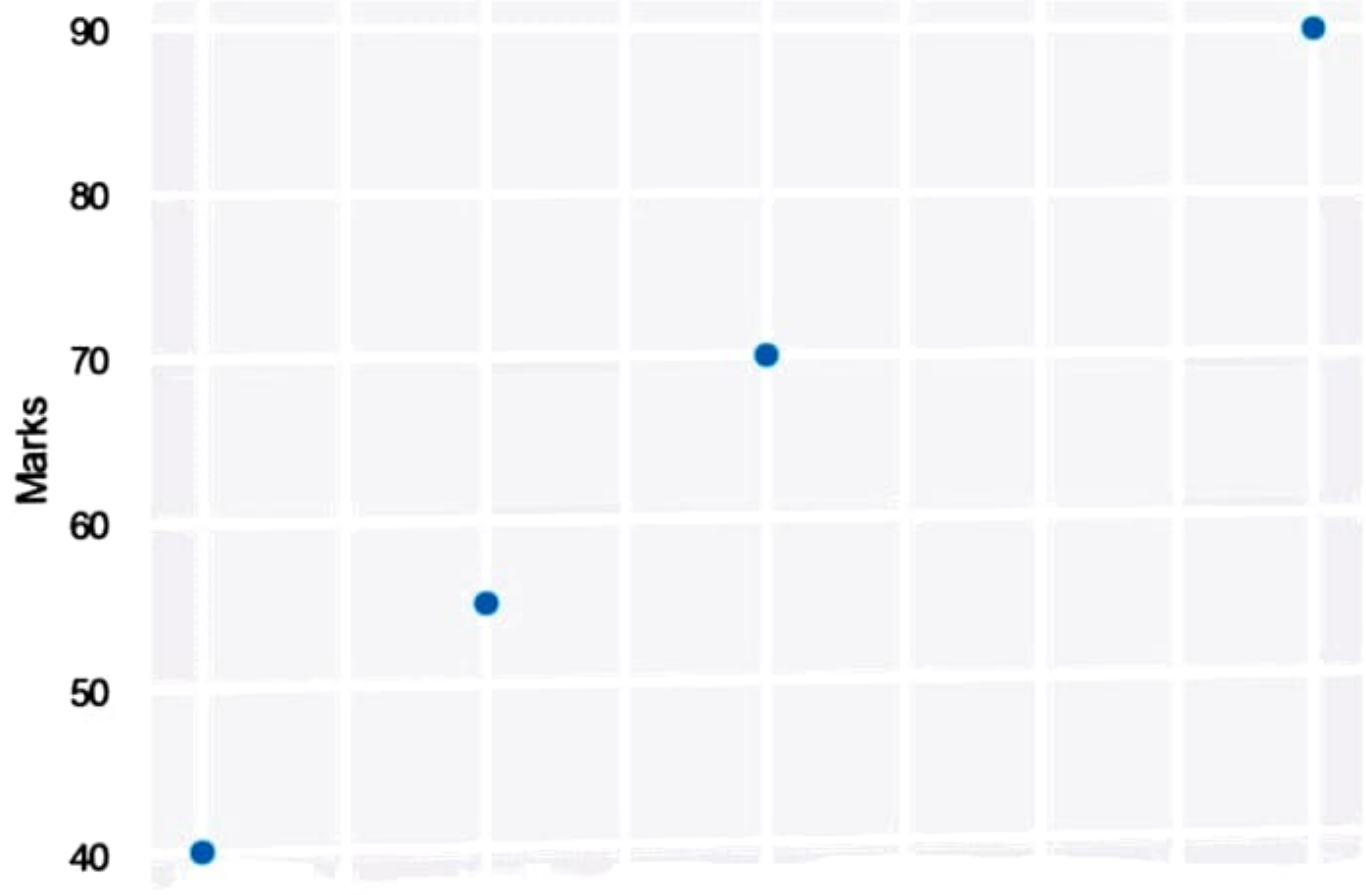
#-Expense distribution

```
import matplotlib.pyplot as plt
labels=['Food','Rent','Travel','Other']
values=[40,30,20,10]
```

```
plt.pie(
    values,
    labels=labels,
    autopct='%1.1f%%'
)
```

```
plt.title("Monthly expenses")
```

```
plt.show()
```

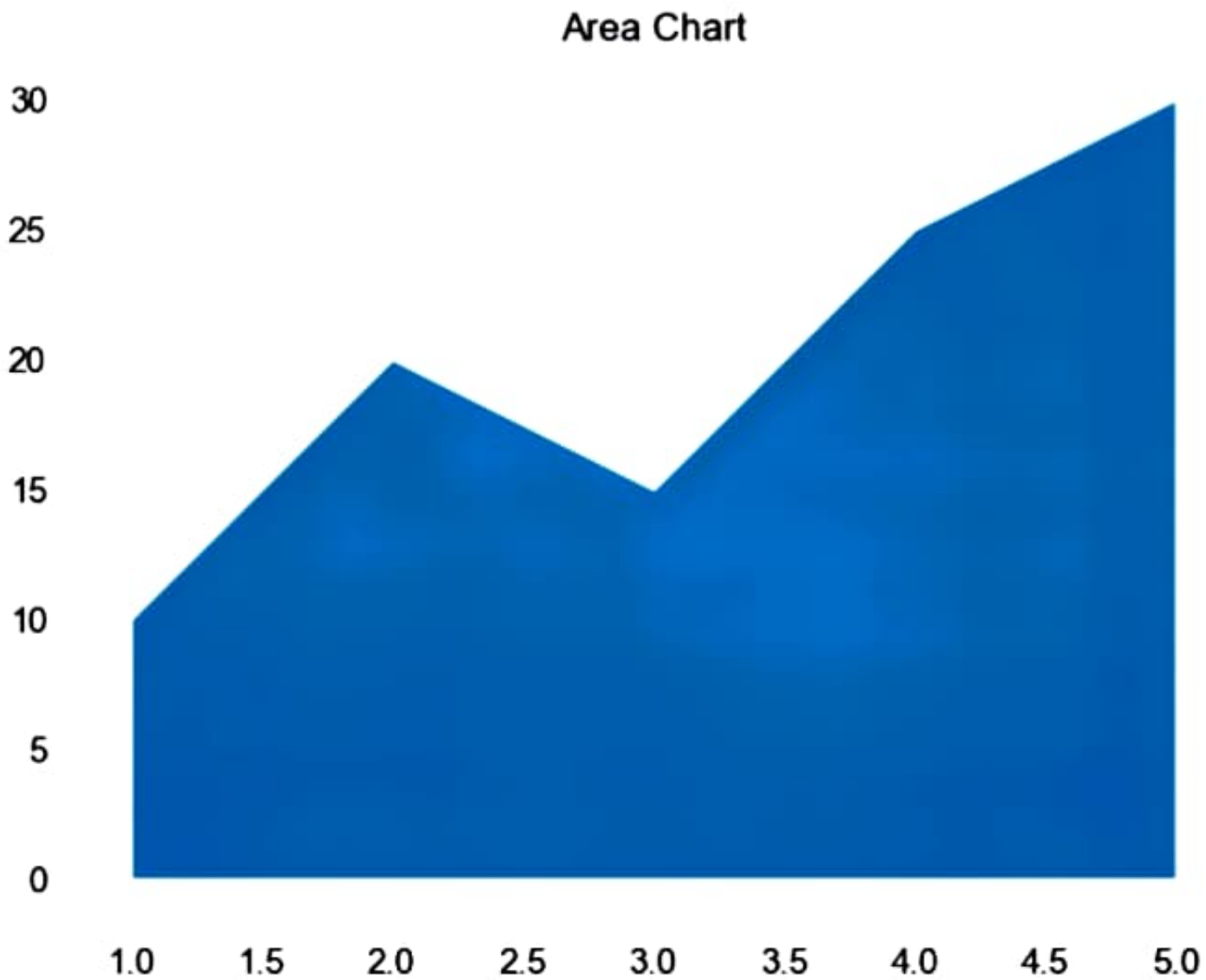


```
import matplotlib.pyplot as plt

x=[1,2,3,4,5]
y=[10,20,15,25,30]

plt.fill_between(x,y)

plt.title("Area Chart")
plt.show()
```



```

import matplotlib.pyplot as plt

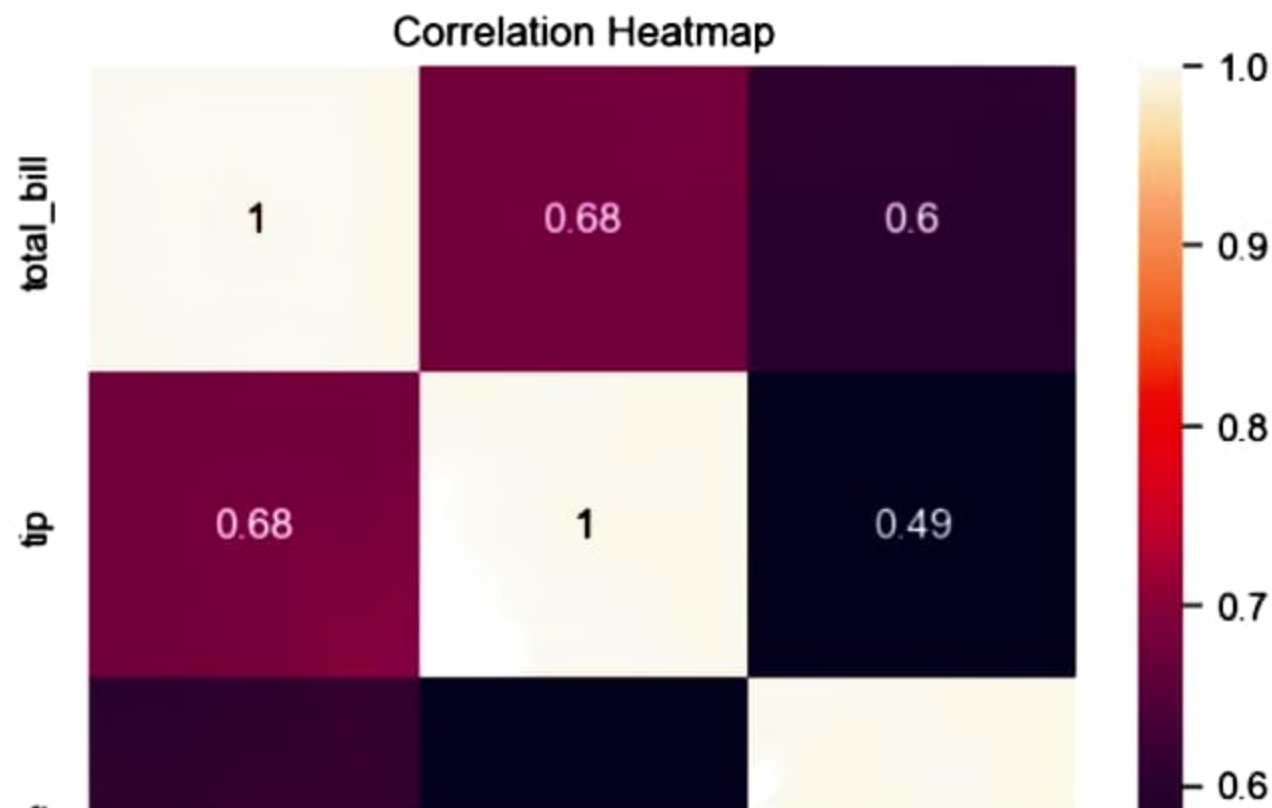
#Load dataset
tips=sns.load_dataset("tips")

#Correlation matrix
corr=tips.corr(numeric_only=True)

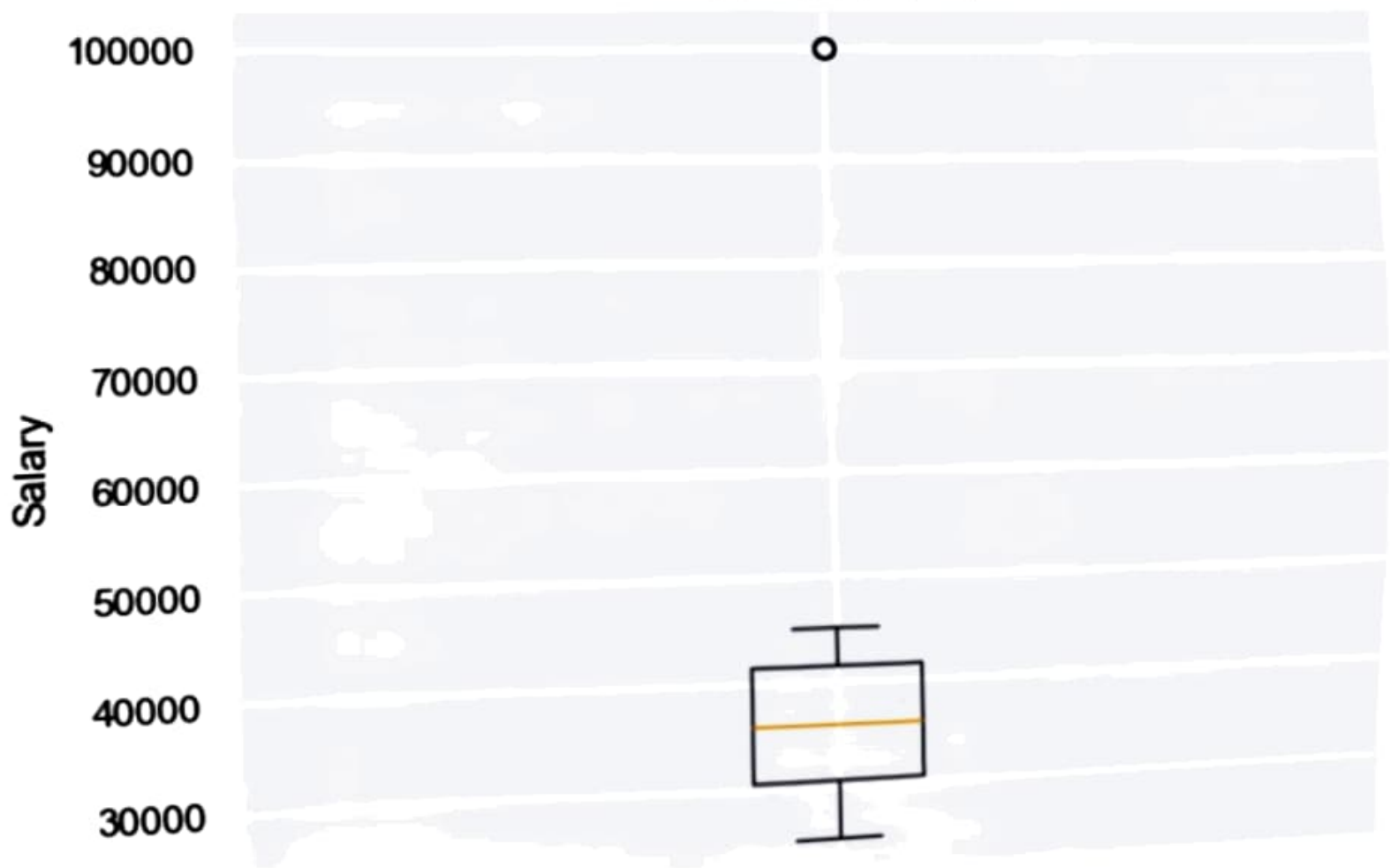
#Heatmap
sns.heatmap(corr,annot=True)\

plt.title("Correlation Heatmap")
plt.show()

```

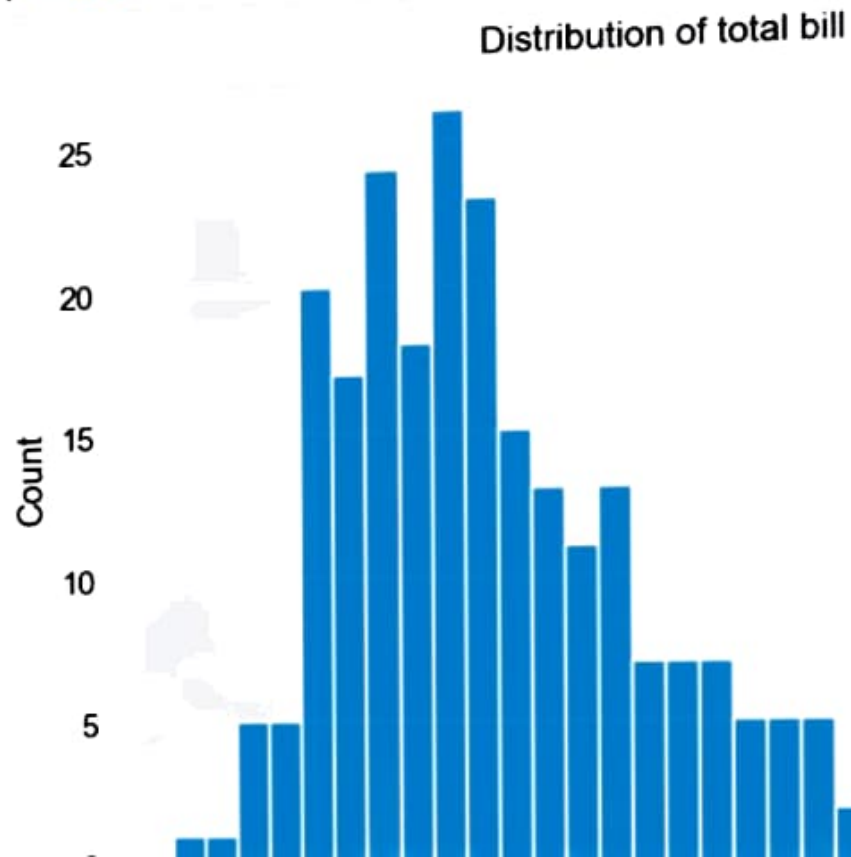


Salary Distribution

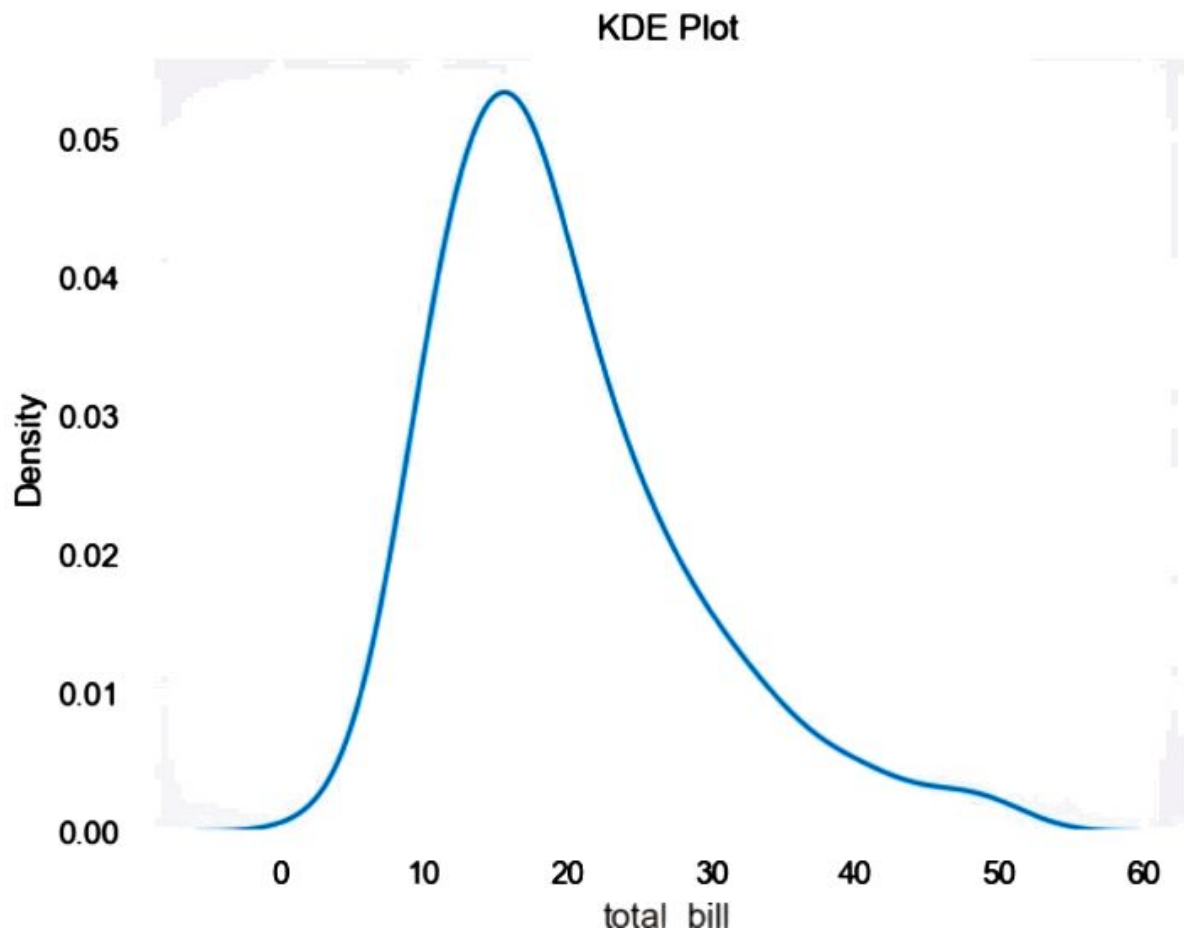


```
#Histogram using seaborn
sns.histplot(
    data=tips,
    x="total_bill",
    bins=30 #bins for accurate without bins it shows apprx val
)

plt.title("Distribution of total bill")
plt.show()
```




```
sns.kdeplot(  
    data=tips,  
    x="total_bill"  
)  
  
plt.title("KDE Plot")  
plt.show()
```

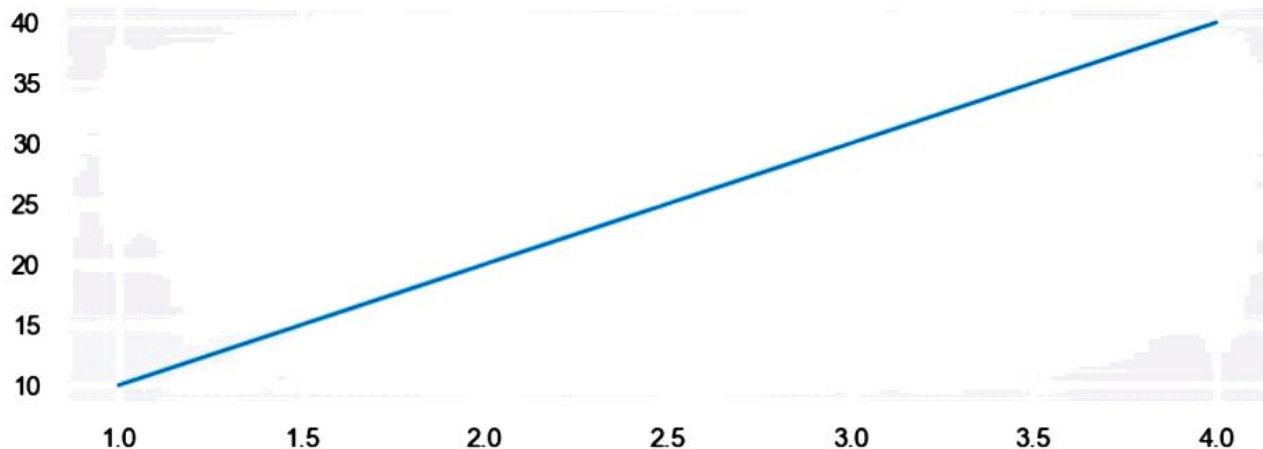


total_bill

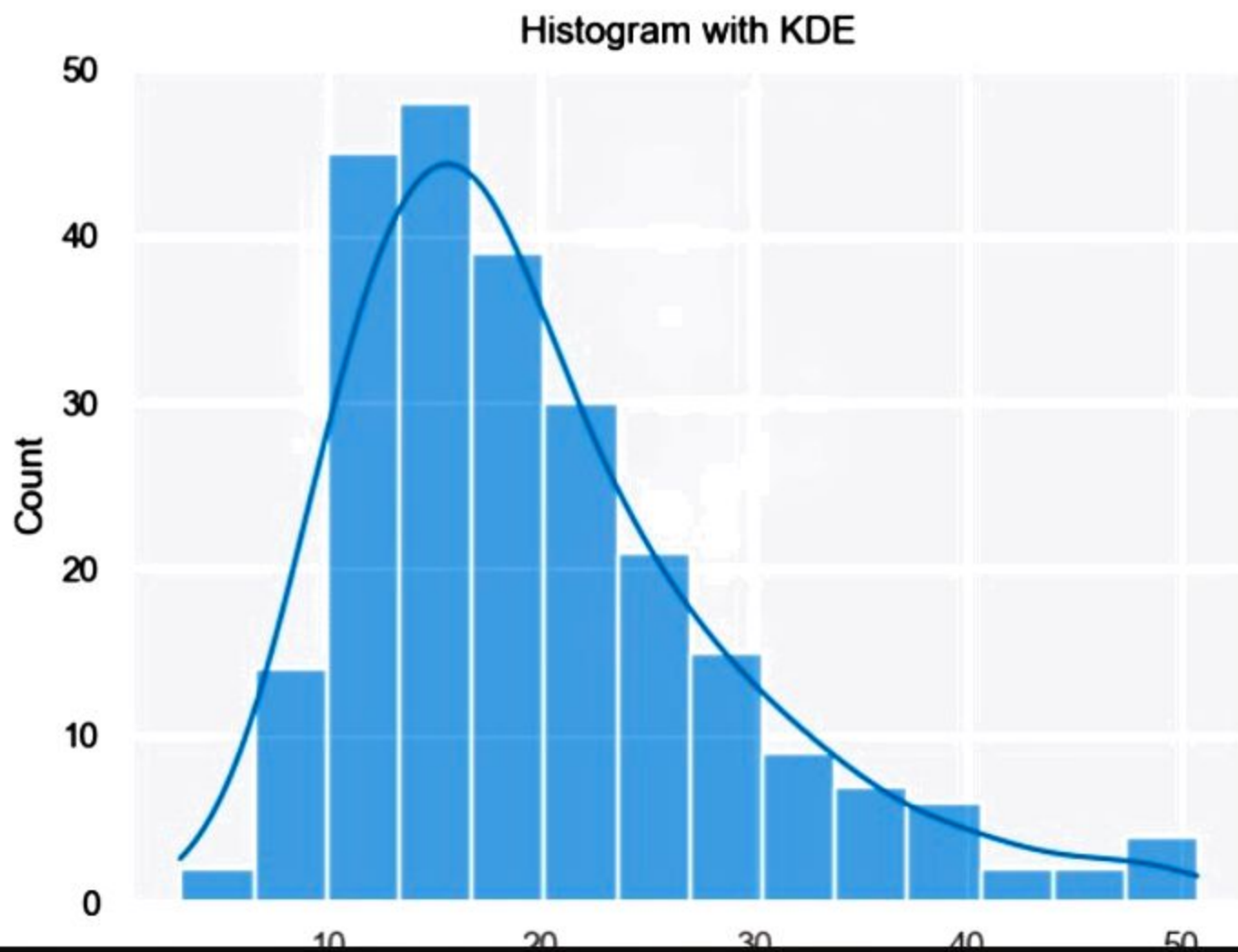
```
[65]: #figure sizing .What is figure size?  
      #it controls the width and height of a chart
```

```
import matplotlib.pyplot as plt  
plt.figure(figsize=(9,3))  
plt.plot([1,2,3,4],[10,20,30,40])  
plt.title("Sales trend")  
plt.show()
```

Sales trend



```
sns.histplot(  
    data=tips,  
    x="total_bill",  
    kde=True  
)  
plt.title("Histogram with KDE")  
plt.show()
```



#Subplots

#it allows multiple charts in a single figure.syntax:plt.subplot(rows,columns,position)

```
import matplotlib.pyplot as plt
```

```
plt.figure(figsize=(8,4))
```

First Plot

```
plt.subplot(1,2,2)
```

```
plt.plot([1,2,3], [10,20,30])
```

```
plt.title("Line Chart")
```

Second Plot

```
plt.subplot(1,2,1)
```

```
(plt.bar)(['A','B','C'], [5,8,6])
```

```
plt.title("Bar Chart")
```

```
(plt.show)
```

```
plt.title("Line")

plt.subplot(2,2,2)

(plt.bar)(['A','B','C'], [5,8,6])

plt.title("Bar")

plt.subplot(2,2,3)

plt.scatter([1,2,3], [4,8,6])

plt.title("Scatter")

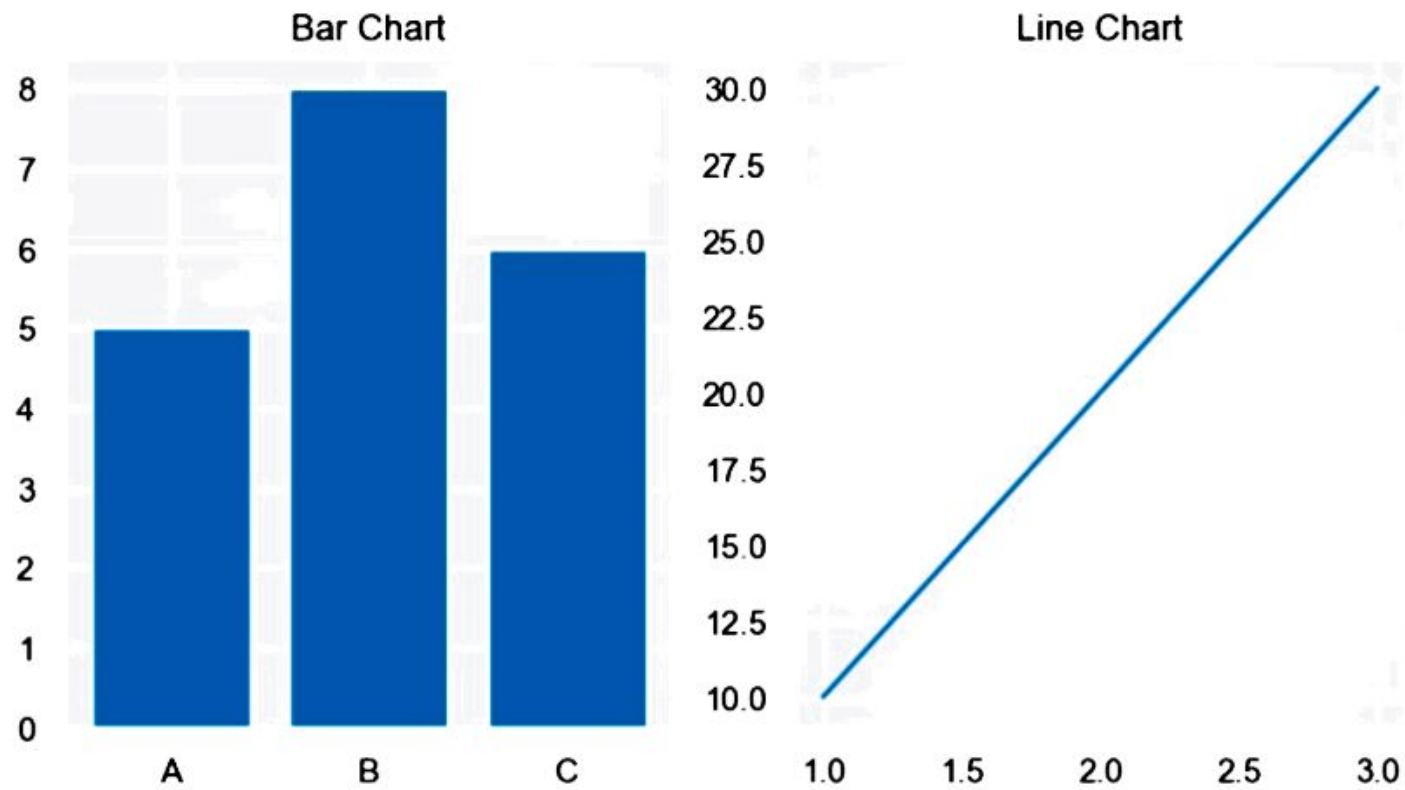
plt.subplot(2,2,4)

plt.pie([30,40,30], labels=['A','B','C'])

plt.title("Pie")
plt.tight_layout()

(plt.show)
```

```
[68]: <function matplotlib.pyplot.show(close=None, block=None)>
```



```
[70]: #four charts
import matplotlib.pyplot as plt

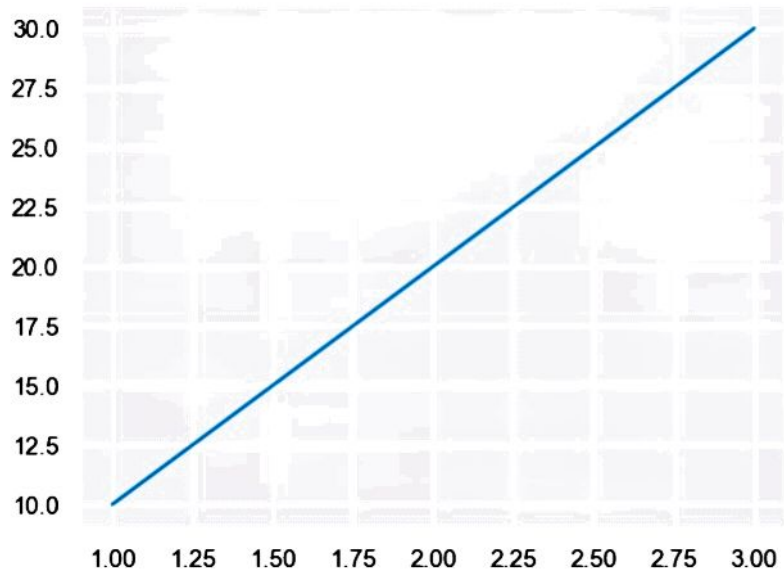
plt.figure(figsize=(10,8))

plt.subplot(2,2,1)

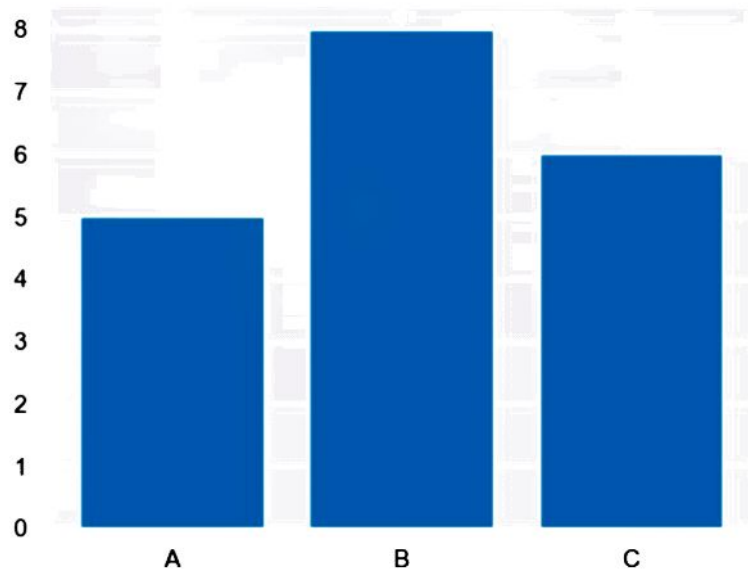
plt.plot([1,2,3], [10,20,30])

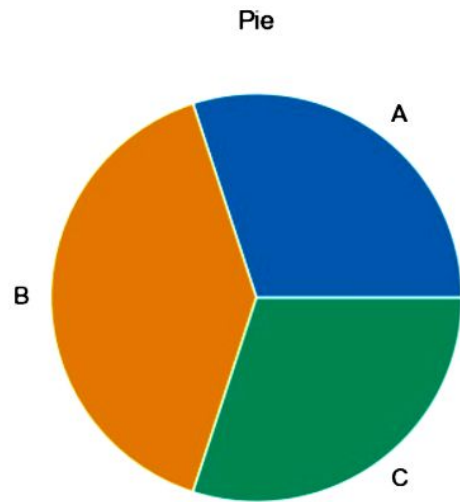
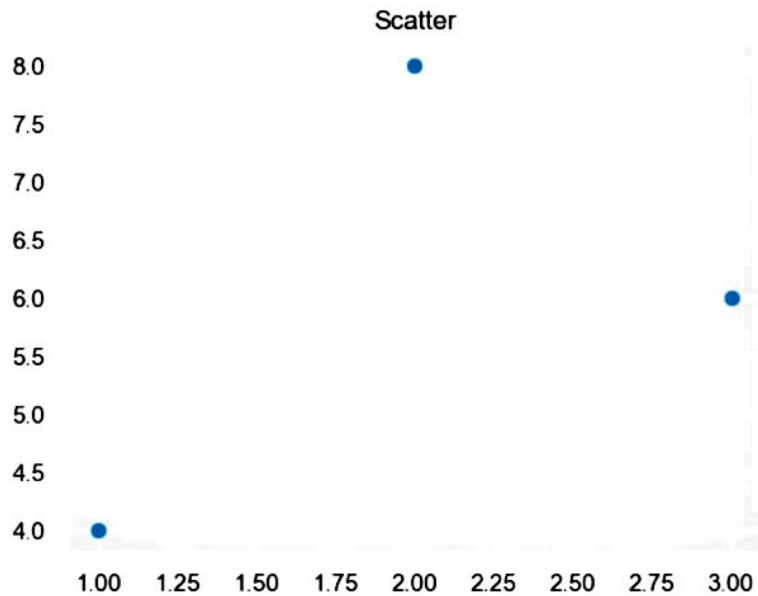
plt.title("Line")
```

Line



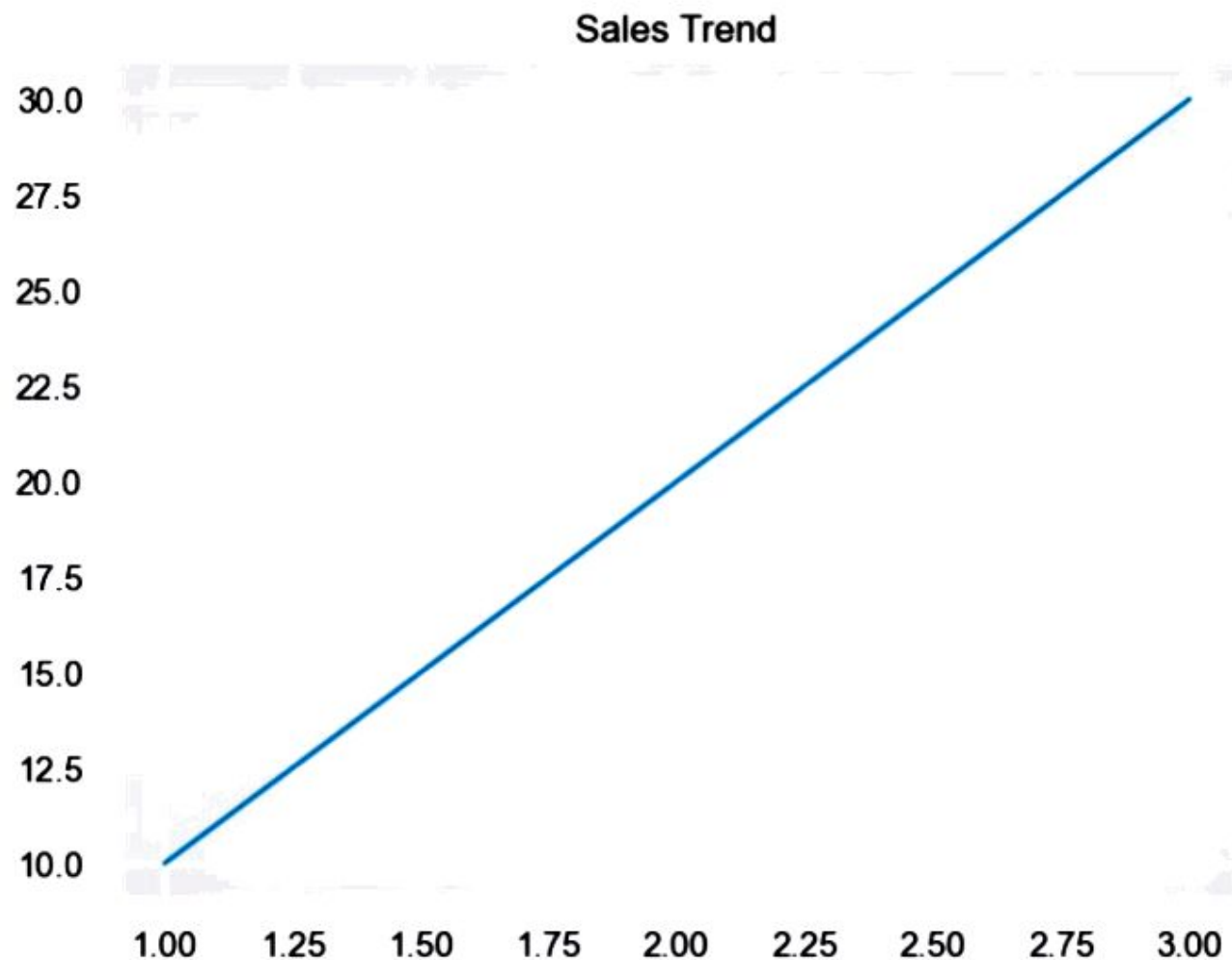
Bar





#plt.savefig() used to save charts as image files

```
import matplotlib.pyplot as plt  
plt.plot([1,2,3],[10,20,30])  
plt.title("Sales Trend")  
plt.savefig("sales_chart.png")  
plt.show()
```



#save with high quality

```
#save with high quality
```

```
plt.savefig(
```

```
    "chart.png",
```

```
    dpi=300
```

```
)
```